substance which is meant to be transmuted.

The disadvantage of this apparatus is that it is extremely expensive. Therefore efforts have been made to achieve the same results by simpler means. In this connection the *Van de Graaff high-voltage generator* deserves attention. It is based on the old, and now hardly ever used principle of the influence machine. This generator consists of a large hollow metal sphere (Figure 35), or cylinder, serving as conductor, with a pulley within it and another pulley underneath it. A wide closed belt of some insulating material, for instance, silk, travels on these two pulleys. Outside of the conductor, electric charge is sprayed on the belt by means of a rectifier and a corona comb, and the belt, carrying this charge, enters the conductor, in which the charge is removed by a second corona comb and is transferred to the conductor. The conductor can thus be charged up to any desired voltage. Certain limitations are imposed by the dimensions of the space in which the generator is installed, since when a certain voltage is reached (the magnitude of which depends on the dimensions of this space and of the conductor itself) a spark jumps across to the walls and discharges the conductor. In 1939, there existed as yet no operating generator capable of attaining more than 2,000,000 volts. Figure 36 shows the largest installation of this nature, which was under construction in the United States several years ago. It is designed to produce 5,000,000 volts, and therefore built in a very large space, an old airship hangar. It has two conductors, supposed to be charged with opposite charges, in order to produce the double voltage for the discharge tube.

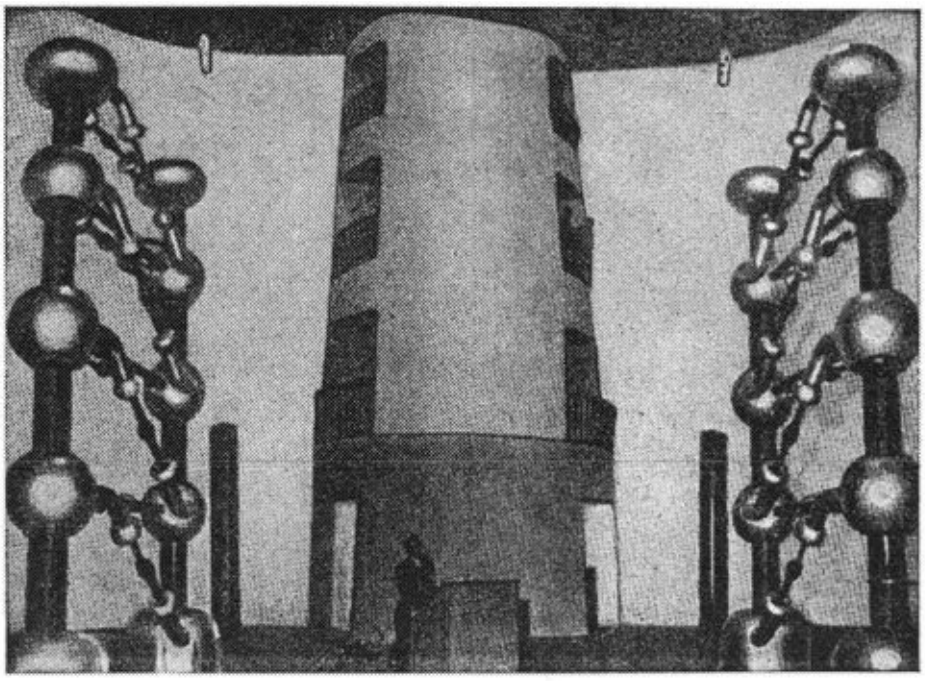


Figure 34.—High tension generator of the Kaiser Wilhelm Institute (Max Planck Institute) in Berlin-Dahlem.